

Paperclip Gateway Final Integration Report

Generated: 2026-05-07

Executive Summary

Paperclip is integrated into Mission Control Gateway as a read-only, policy-gated Workforce Operations Layer. It is visible to Gateway, Agent Zero, Hermes, Pi, and SpaceAgent through registry and bridge contracts. Paperclip is not the commander, does not replace existing agents, and does not get execution authority by default.

Final status: PARTIAL GO for read-only Gateway visibility and route planning. NO-GO for live Paperclip workforce operations until the sandbox service/UI is running and its health endpoint passes.

No external writes were executed. No Zapier, HeyGen, SMB, farmer, email, Drive, or OneDrive action was run. No secrets were exposed or committed.

Phase 291-300 Results

Phase	Result	Evidence
291	Blocked	Paperclip user and system services are inactive.
292	Blocked	No listener or health response on the expected Paperclip port.
293	Blocked	Loopback and Tailnet health checks returned connection refused/no HTTP response.
294	Passed	Gateway registry contains paperclip as Paperclip Workforce Control Plane with blocked service status.
295	Passed	Agent Zero bridge context exposes Paperclip status/task routes, with writes blocked.
296	Passed	Hermes bridge context exposes Paperclip registry and planning-only proposal surface.
297	Passed	Pi dispatcher recommendation payload can route advisory Paperclip workforce decisions.
298	Passed	SpaceAgent appears in Paperclip research-task handoff and work-product flow, storage blocked until adapter/session.
299	Passed	1,000 deterministic scenarios passed with zero writes, zero secrets, and no fake completion.
300	Completed	Markdown and PDF report artifacts created in Mission Control runtime reports.

What Was Implemented

- Added a final Paperclip Gateway routing gauntlet covering 1,000 deterministic scenarios.
- Verified Paperclip remains subordinate to Gateway and Agent Zero.
- Verified Hermes can create Paperclip proposal payloads only in planning mode.
- Verified Pi can recommend Paperclip routes only in shadow/advisory mode.
- Verified SpaceAgent can be represented in Paperclip research-task tracking without executing writes.
- Verified co-worker definitions, co-worker policy checks, workforce flows, task handoffs, proposals, research tasks, and token-governor dry runs remain read-only or session-gated.
- Created this final Markdown/PDF report.

What Stayed Blocked

- Paperclip sandbox service/UI is not running.
- Paperclip local health endpoint is not reachable.
- Paperclip Tailnet UI is not reachable.
- Owner login to Paperclip UI is not proven in this final block.
- Paperclip live issue/task/work-product write adapter is not configured.
- Persistent production service is not enabled.
- Live Paperclip workforce operations remain blocked until sandbox health and auth are proven.

Gateway Roles

- Owner: final authority.
- Gateway: routing, policy, registry, audit, documentation, and memory hub.
- Agent Zero: commander and final decision-maker.
- Hermes: lieutenant and workflow/skill builder.
- Pi: dispatcher candidate and route optimizer in shadow/advisory mode.
- SpaceAgent: web/browser/YouTube/Firecrawl research specialist.
- Paperclip: workforce/company/task orchestration layer, subordinate to Gateway and Agent Zero.
- OpenCloud: retained worker/runtime engine.
- OpenClaw+: retained runtime, skills, adapters, and reports layer.
- Existing agents: retained specialist workforce.

Security and Governance

- Paperclip routes are protected by Mission Control auth.
- Unauthenticated Paperclip bridge routes are covered by tests returning 401/403.
- External writes remain disabled without Bridge Session and scoped adapter approval.
- Paperclip co-workers cannot get direct secret access, raw root shell, Docker socket, or self-promotion.
- Paperclip write flows return exact blockers instead of fake completion.
- Owner-facing payloads are checked for no local path leaks, no token patterns, no secret placeholders, and no fake Done.

Validation

Mission Control validation passed:

- git diff --check: passed.
- pnpm run typecheck: passed.
- pnpm run build: passed.
- pnpm test: passed, 131 test files and 1,230 tests.
- Focused Paperclip tests: passed, 42 tests across bridge, routes, and final gauntlet.
- New Paperclip gauntlet: passed, 1,000 scenarios, 0 failures.

Service checks:

- Mission Control service: active.
- ClaudeClaw service: active.
- Hermes gateway service: active.
- Build-Wiki/Farmer timer: active.
- Agent Zero container: running.

- Paperclip service: inactive/blocking live UI and health.

Dirty Tree Notes

The Mission Control repo already had many unrelated untracked runtime/design artifacts before this closure work. This phase only stages the Paperclip final gauntlet and Paperclip final report artifacts.

Commits

Existing Paperclip commits already pushed:

- cd18aec - feat(gateway): add paperclip workforce status panel
- dfb65a4 - feat(gateway): add paperclip workforce flow contract
- c8ca7e9 - test(gateway): harden paperclip bridge routes

The final closure commit is the commit that contains this report and the Paperclip gauntlet proof.

Rollback

Rollback the final Paperclip closure commit with:

```
bash
git revert <final-paperclip-closure-commit>
```

Rollback the earlier Paperclip Gateway work, if needed, in reverse order:

```
bash
git revert c8ca7e9
git revert dfb65a4
git revert cd18aec
```

Exact Next Step

Start Paperclip in a sandbox-only localhost/Tailnet mode, prove /api/health, prove owner login, then rerun the Paperclip status/UI smoke before enabling any persistent service or write adapter.